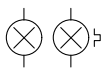
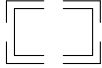
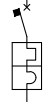
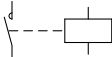
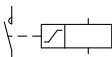
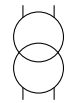
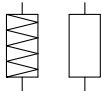


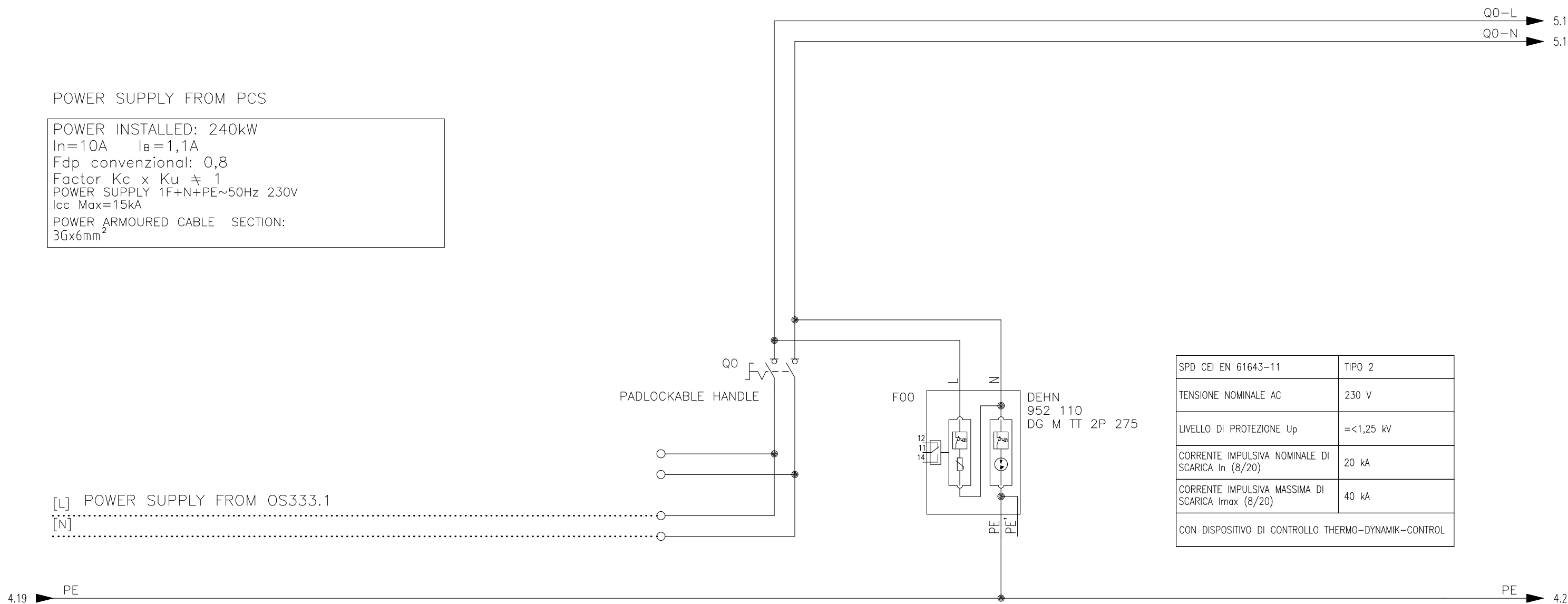
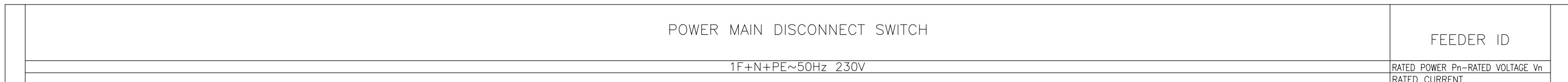
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		
TENDRARA PROJECT GAS TREATMENT AND LNG PRODUCTION PLANT ON RENTAL BASIS																					
SAFETY CRITICAL ELEMENT LIST																					
IFI	1	27/02/2026	AS-BUILT				ADF	GDP	IC												
IFI	0	26/03/2024	ISSUED FOR COMMENTs				ADF	GDP	IC												
Status	Rev. n.	Rev. n.	Rev. n.				Rev. n.	Rev. n.	Rev. n.												
Rev. Index		Date	Description				Prepared	Checked	Approved												
Client							Project Title														
ManaEnergy Managem Group MANAENERGY Mahaj Riad Center, Batiment 6, 3eme etage Avenue Al Arz, Hay Ryad, 10100 Rabat, Morocco							TENDRARA PROJECT														
Contractor							Provision of a m-LNG plant for production for ManaEnergy														
ITALFLUID ITALFLUID GEOENERGY Marocco Branch Boulevard de Bourgogne Rue Jaafar Ibn Habib Residence Al Mashrik II 1ere Etage N3 Casablanca																					
Engineering																					
ITALFLUID ITALFLUID GEOENERGY Via Antonio Vivaldi, 4 Montesilvano (PE) Tel:085-4224374 / fax: 085-4220742																					
Facility Name			Location				Scale		Sheet of Sheets												
TENDRARA			MOROCCO				A4		1 of 14												
Document Title					Doc. Code																
P-PN-OT331.1					ELD01-ING-FR01																
AMMINE COLUMNS					Vendor Doc. code																
					IE24090_SE29_P-PN-OT331.1																
			Project : P–PN–OT331.1				Customer : ManaEnergy Managem Group				Object :										
General Contractor :		Manufacture : esa esaelettromeccanica.com		AMMINE COLUMNS				Contract : IE24090				Rev.		Data :				By :			
								S/N : IE24090_SE29				Rev. 1		Data : 27/02/2026				Date : 26/03/2024			
														Date : 26/03/2024							
														Sheet : 1				of : 14			

CIRCUIT SYMBOLS ACCORDING TO CEI STANDARD

	ISOLATOR		ELECTRONIC TIMER WITH PICK-UP DELAY		MUSHROOM PUSH-BUTTON PULSE CONTACT WITH STAY-PUT MAKE CONTACT OPERATED BY STRIKING		STEDY OR FLASHING LIGHT
	FUSED ISOLATOR		ELECTRIC PULSE METER		KEY OPERATING DEVICE FOR SELECTORS, PUSH-BUTTON, ETC.		PLC UNIT
	CIRCUIT BREAKER WITH THERMAL AED MAGNETIC OVERCURRENT RELEASES		FUSE CARRIER		PRESSURE SWITCH		CONTROL INSTRUMENT
	CIRCUIT BREAKER WITH MAGNETIC RELEASE		MAKE CONTACT OF CONTACTOR		TEMPERATURE SWITCH		ENERGY METER
	CIRCUIT BREAKER WITH THERMAL, MAGNETIC AND DIFFERENTIAL RELEASES		MAKE CONTACT OF CONTACTOR		FLOW SWITCH		MEASURING INSTRUMENT
	RESIDUAL CURRENT CIRCUIT BREAKER "AC" TYPE FOR ALTERNATING CURRENT "A" TYPE FOR ALTERNATING OR PULSATING CURRENTS		MAKE AND BREAK CONTACT		FLOAT SWITCH		VOLTAGE TRANSFORMER
	THERMAL RELAY (OVERLOAD RELAY)		MAKE AND BREAK CONTACT WITH DROP-OUT DELAY		BUCHHOLZ RELAY		CURRENT TRANSFORMER
	COIL OF CONTACTOR RELAY		MAKE AND BREAK CONTACT WITH PICK-UP DELAY		CAM OPERATED DEVICE		ACOUSTIC SIGNALLING DEVICE
	COIL OF SHUT-OFF ALARM MAKE CONTACT ON ENERGIZATION		MAKE AND BREAK CONTACT WITH DROP-OUT AND PICK-UP DELAY (FLIP-FLOP)		PROXIMITY SWITCH		RESISTANCE
	COIL OF SHUT-OFF ALARM DENERGIZES IN CASE OF UNDERVOLTAGE		MAKE AND BREAK CONTACT OF THERMAL RELAY		TEMPERATURE SENSOR		CAPACITOR
	COIL OF OPERATING DEVICE WITH DROP-OUT DELAY		MAKE AND BREAK CONTACT OF LIMIT SWITCH		PHOTOELECTRIC SENSOR		SINGLE-PHASE ELECTRIC MOTOR
	COIL OF OPERATING DEVICE WITH PICK-UP DELAY		MULTI POSITION SWITCH (GENERAL SYMBOL)		SOLENOID VALVE		THREE-PHASE ELECTRIC MOTOR
	COIL OF OPERATING DEVICE WITH DROP-OUT AND PICK-UP DELAYS		MAKE AND BREAK CONTACT OPERATED BY TURNING WITH SPRING RETURN		PHOTOCELL		
	COIL OF LATCHING RELAY (STEP-BY-STEP FUNCTION)		MAKE AND BREAK CONTACT OPERATED BY PUSHING WITH SPRING RETURN				
	ELECTRONIC OR ELECTRO-MECHANICAL TIME SWITCH		MAKE AND BREAK CONTACT OPERATED BY PULLING WITH OR WITHUOT SPRING RETURN				

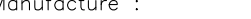
CONDUCTOR COLOR CODING INSIDE CONTROL PANEL	
AC AND DC POWER CIRCUIT	BLACK
NEUTRAL CONDUCTOR OF POWER CIRCUIT	LIGHT–BLUE
PROTECTIVE CONDUCTOR (EARTH CONDUCTOR)	YELLOW–GREEN
110/220V AC AUXILIARY CIRCUIT (*)	RED
24V AC AUXILIARY CIRCUIT	WHITE
24V AC AND DC PLC INPUT	GRAY
24V AC PLC OUTPUT	WHITE
24V AC AND DC INTERLOCKING CIRCUIT	ORANGE
24V DC AUXILIARY CIRCUIT AND PLC OUTPUT	DARK–BLUE
(*) : RED FOR 110V AND BROWN FOR 220V, IN CASE BOTH VOLTAGE ARE USED	

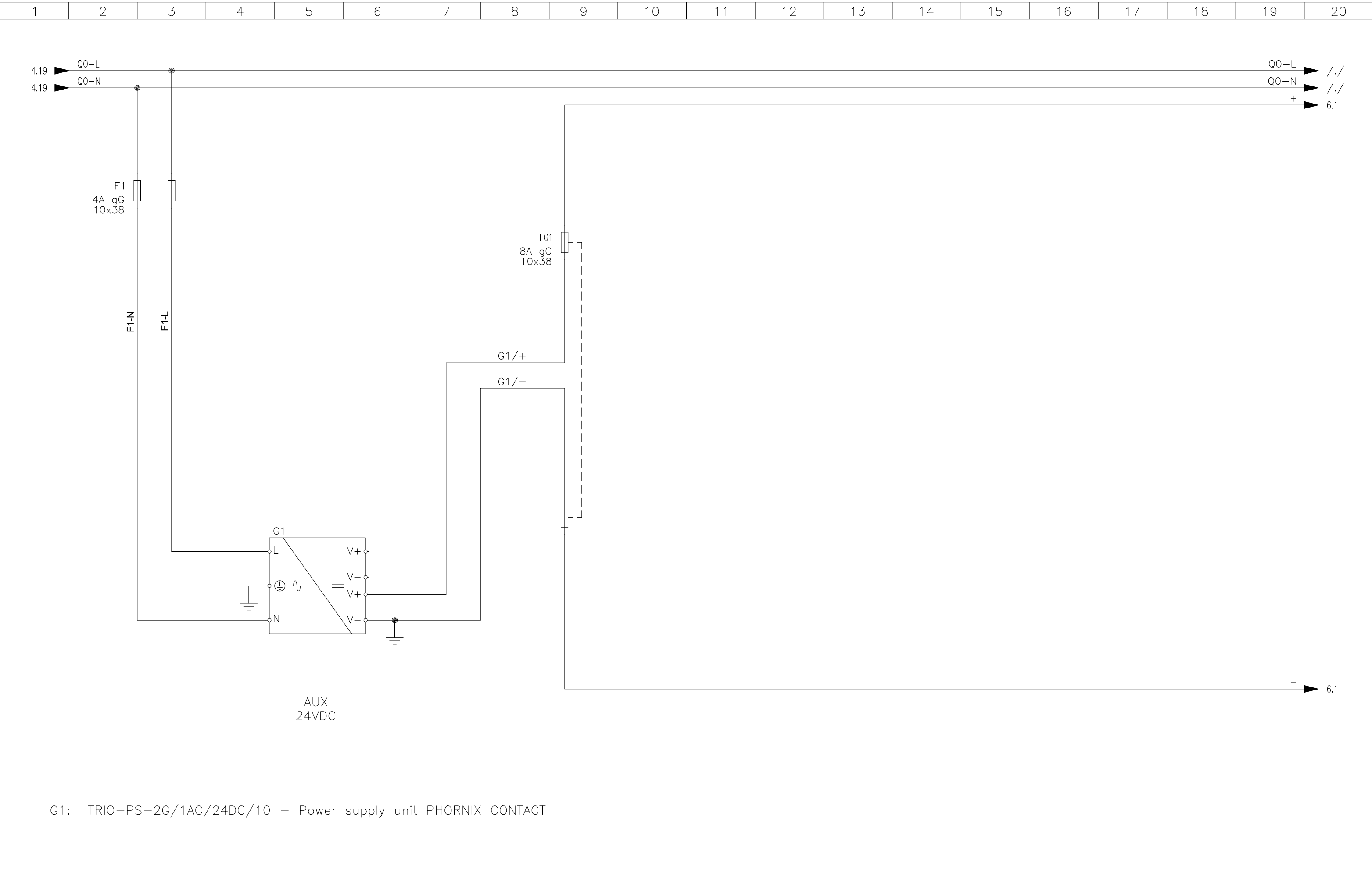
CODE COLOUR	
COLOUR	CODE LETTERS
BLACK	BK
BROWN	BN
RED	RD
ORANGE	OG
YELLOW	YE
GREEN	GN
BLUE	BU
GREY	GY
WHITE	WH
SILVER	SR
CLEAR	CL
GREEN AND YELLOW	GNYE

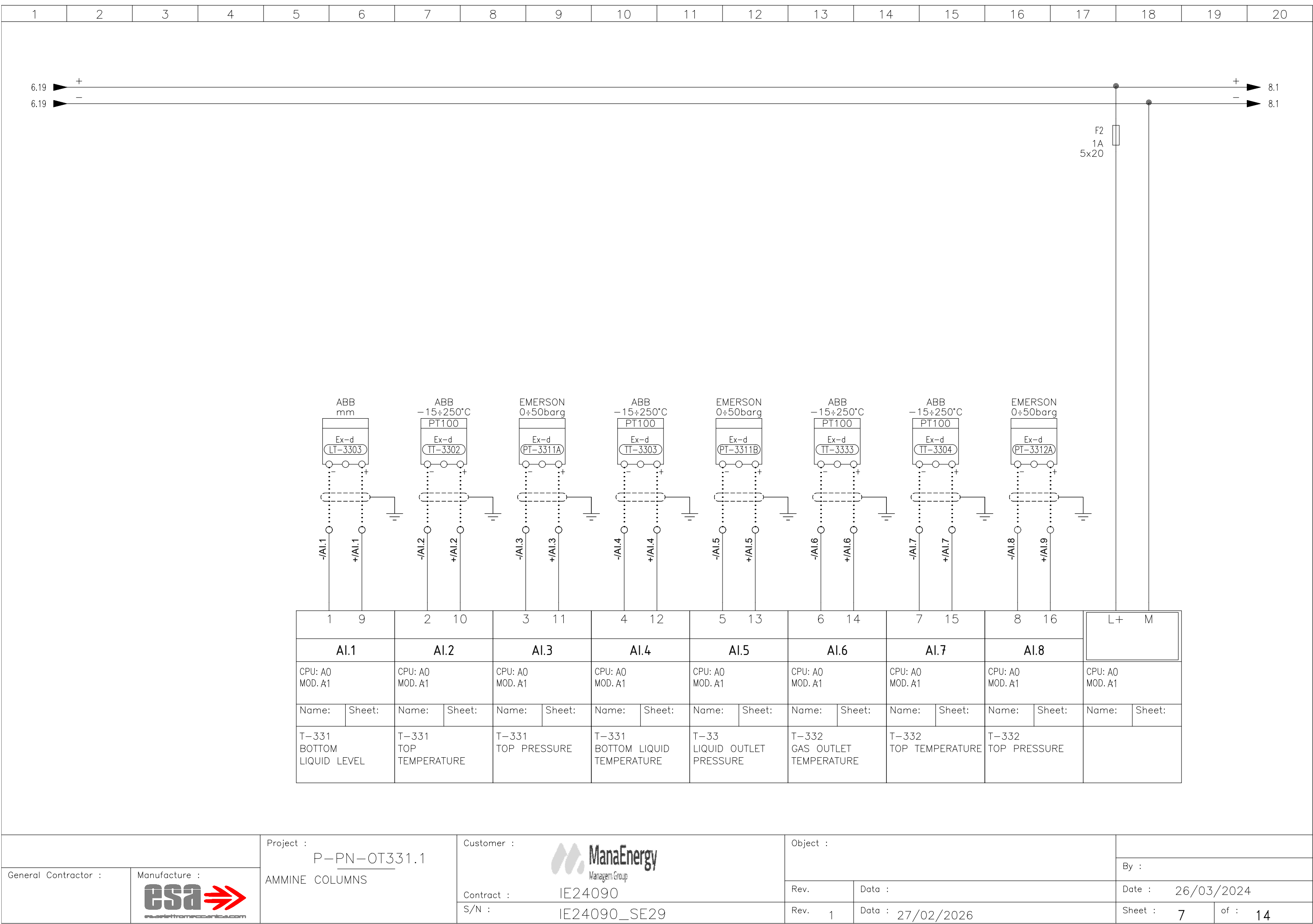


SPD CEI EN 61643-11	TIPO 2
TENSIONE NOMINALE AC	230 V
LIVELLO DI PROTEZIONE Up	=<1,25 kV
CORRENTE IMPULSIVA NOMINALE DI SCARICA In (8/20)	20 kA
CORRENTE IMPULSIVA MASSIMA DI SCARICA I _{max} (8/20)	40 kA
CON DISPOSITIVO DI CONTROLLO THERMO-DYNAMIK-CONTROL	

PROTECTION	TECHNOR PSRC-220	TYPE	PROTECTION
	20A	RATED BREAKING CAPACITY I _{cu}	
		RATED or THERMAL CURRENT I _n or I _{th}	
		MAGNETIC RELEASE CHARACTERISTICS I _m	
		RATED R.C. SENSITIVITY I _{dn}	
CONDUCTOR		WIRE SECTION	CONDUCTOR
		CABLE TYPE	
		CABLE CARRYING LOAD I _z	
		CABLE LENGTH	
		LINE ID	
		WIRE ID ON TERMINAL	

General Contractor :		Manufacture : 	Project : P-PN-OT331.1 AMMINE COLUMNS	Customer : 	Object :		By :	
					Rev.	Data :	Date : 26/03/2024	
			Contract : IE24090	S/N : IE24090_SE29	Rev. 1	Data : 27/02/2026	Sheet : 4	of : 14





ABB

-15÷250°C

PT100

Ex-d

TT-3333

-

+

-

+

-

+

ABB

-15÷250°C

PT100

Ex-d

TT-3304

-

+

-

+

-

+

EMERSON

0÷50barg

Ex-d

PT-3312A

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16

L+

M

AI.1

AI.2

AI.3

AI.4

AI.5

AI.6

AI.7

AI.8

CPU: A0

MOD. A1

CPU: A0

MOD. A1

CPU: A0

MOD. A1

CPU: A0

MOD. A1

CPU: A0

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Sheet:

T-331

BOTTOM

LIQUID LEVEL

T-331

TOP

TEMPERATURE

T-331

TOP PRESSURE

T-331

BOTTOM LIQUID

TEMPERATURE

T-33

LIQUID OUTLET

PRESSURE

T-332

GAS OUTLET

TEMPERATURE

T-332

TOP TEMPERATURE

T-332

TOP PRESSURE

General Contractor :

Manufacture :
esa
esa-electronic.com

Project :
P-PN-OT331.1
AMMINE COLUMNS

Customer :

Contract : IE24090
S/N : IE24090_SE29

Object :

Rev.

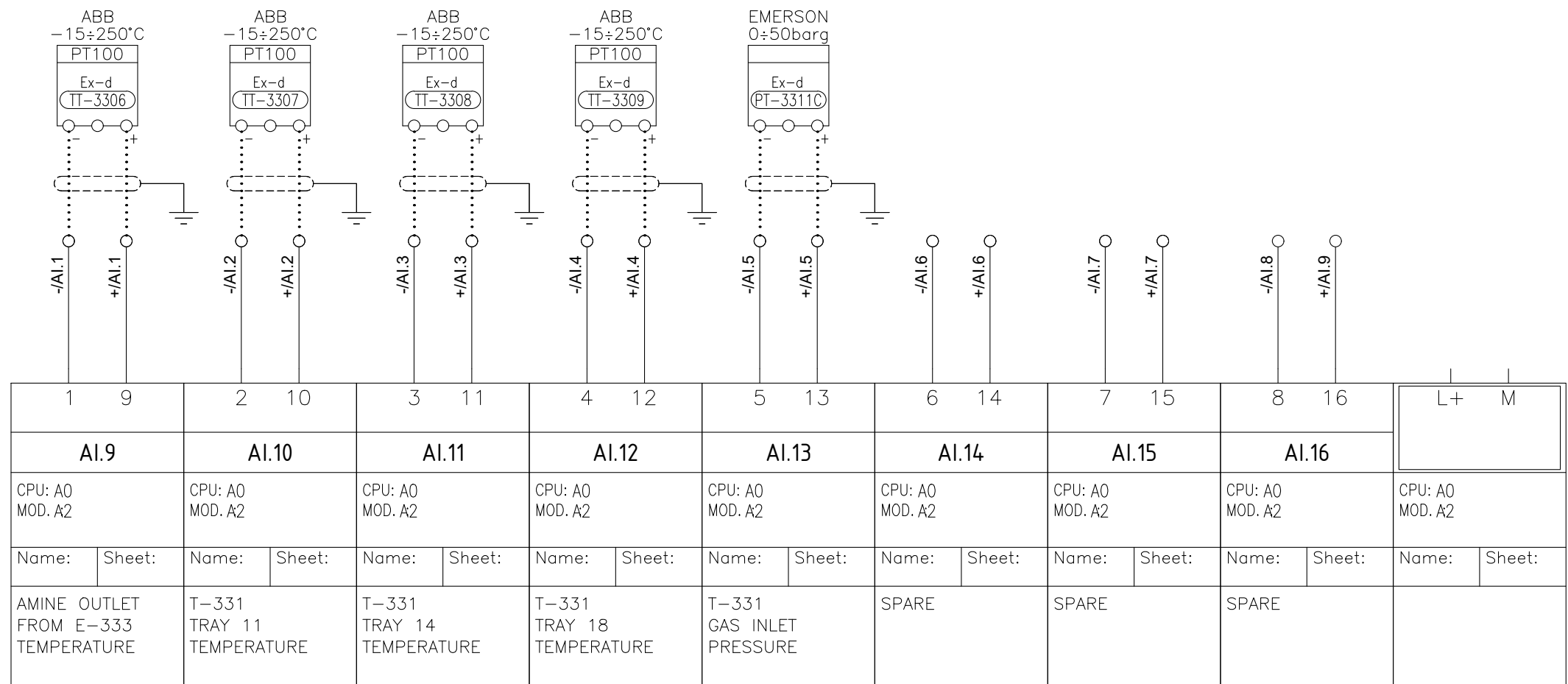
Data :

Rev.

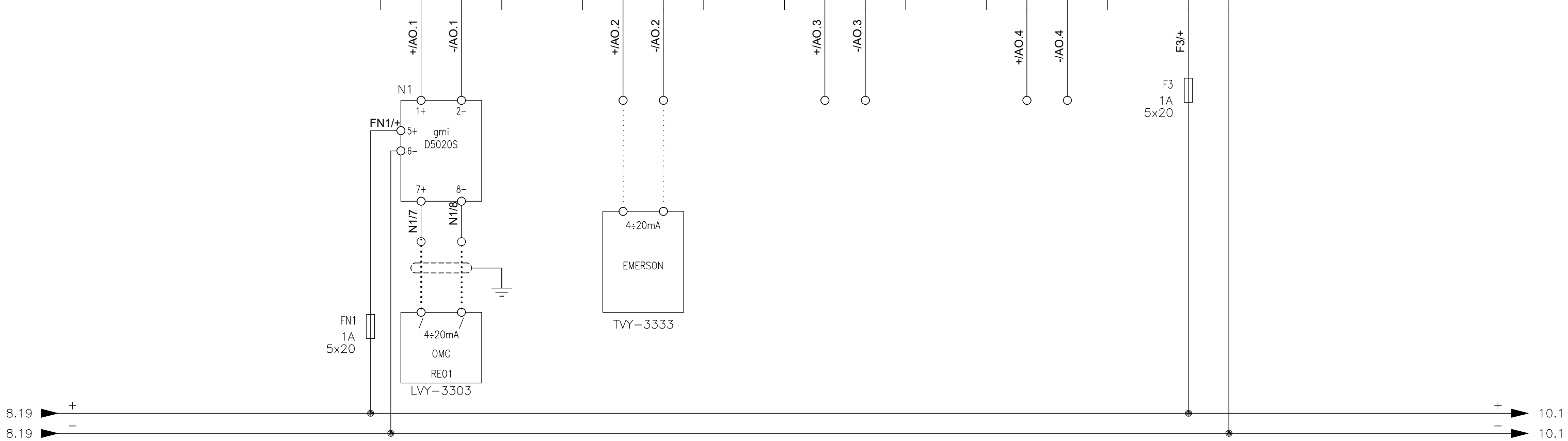
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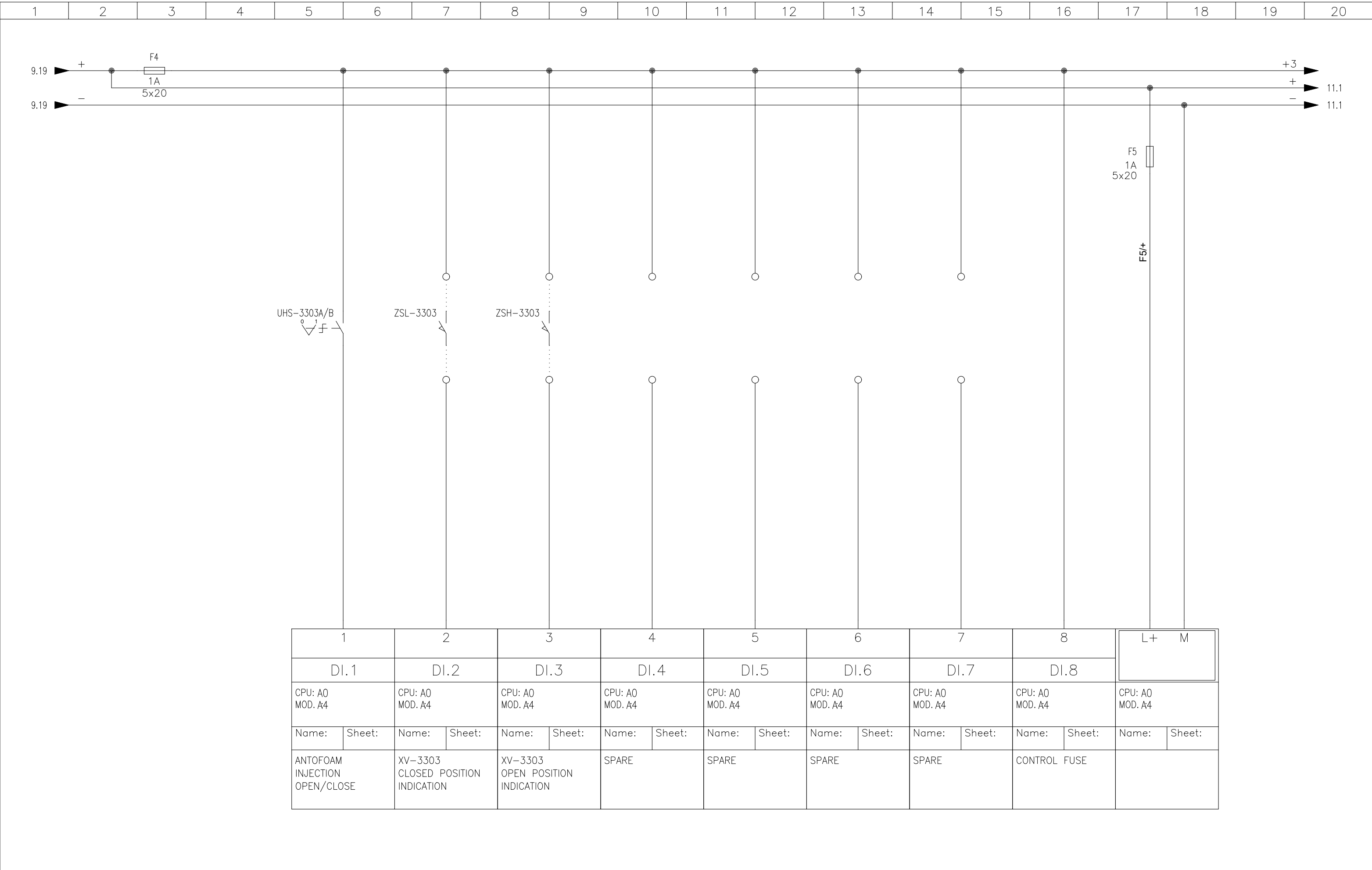
Data : 27/02/2026

By :
Date : 26/03/2024
Sheet : 7 of : 14

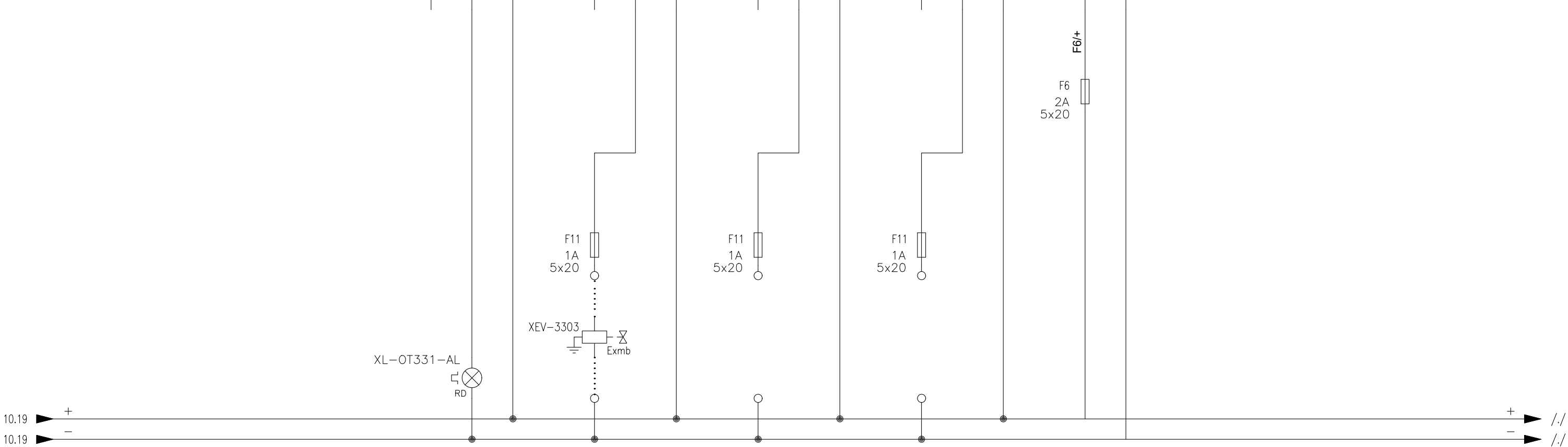


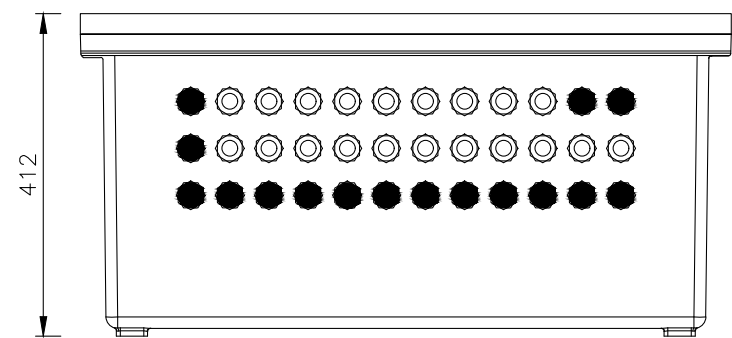
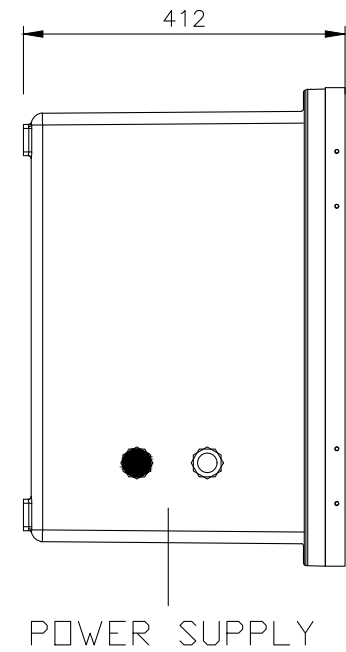
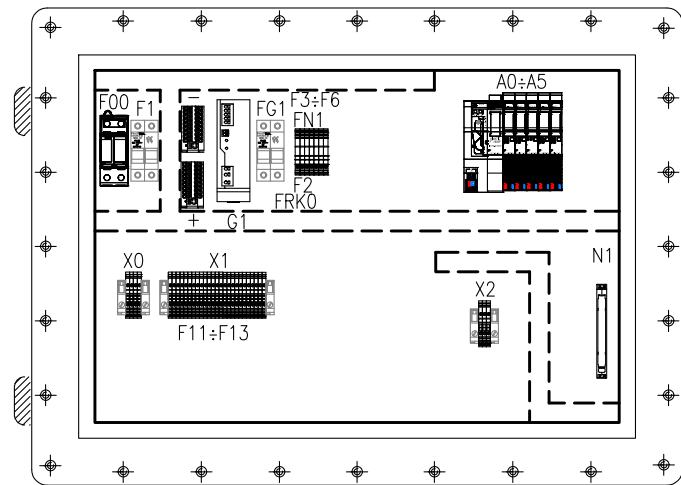
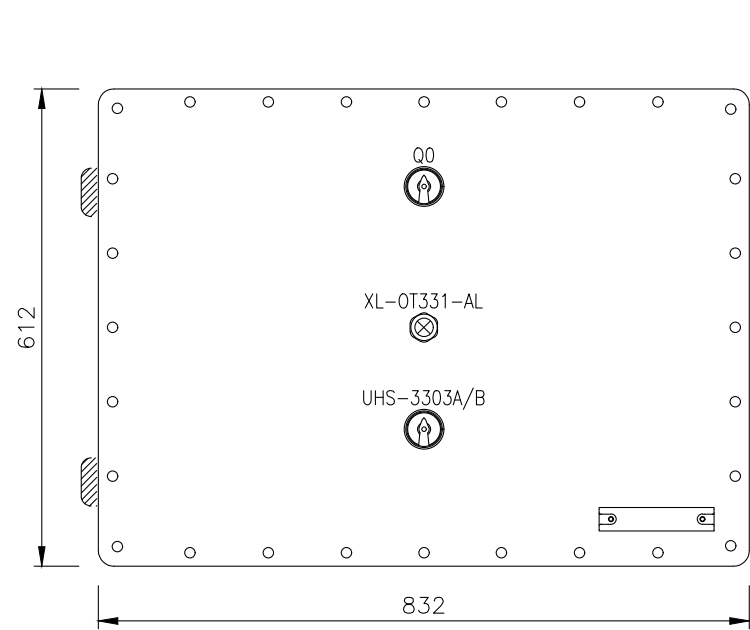
T-331 BOTTOM LIQUID LEVEL CONTROL				T-332 TEMPERATURE CONTROL				SPARE				SPARE							
Name:		Sheet:		Name:		Sheet:		Name:		Sheet:		Name:		Sheet:		Name:		Sheet:	
CPU: A0 MOD. A3				CPU: A0 MOD. A3				CPU: A0 MOD. A3				CPU: A0 MOD. A3				CPU: A0 MOD. A3			
A0.1				A0.2				A0.3				A0.4				L+M			
91513				102614				113715				124816							





PACKAGE ALARM OR TRIP FLUSCHING MEANS SOLID MEANS TRIP			ANTIFOAM INJECTION TO T-332			SPARE			SPARE				
Name:		Sheet:	Name:		Sheet:	Name:		Sheet:	Name:		Sheet:	Name:	Sheet:
CPU: A0 MOD. A5			CPU: A0 MOD. A5			CPU: A0 MOD. A5			CPU: A0 MOD. A5			CPU: A0 MOD. A5	
D0.1			D0.2			D0.3			D0.4			L+M	
1	5	9	2	6	10	3	7	11	4	8	12		





MILANO-ITALY Tel 024455358 Fax 0248401741

Type: EJB-H
230 Vac 50 Hz 16 Å
Max Dis. P: 53 W
T. Cable
Year s.n 25/CMC/0180

CE 0722 Ex II 2GD
INERIS18ATEX0026X/00
Ex db [ia Ga] IIB T6
Ex tb [ia Da] IIC T85 °C IP66
-20°C < Tamb < +50°C

Cable entries: see instructions
WARNINGS: DO NOT OPEN WHILE ENERGIZED
DO NOT OPEN IF AN EXPLOSIVE ATMOSPHERE MAY BE PRESENT
READ INSTRUCTIONS NOTE BEFORE INSTALLATION AND USE

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
INITIALS	SUPPLIER PN	DESCRIPTION							SUPPLIER										
–	3273110	PTFIX 6/18X2,5–NS35 GY – Distribution block							PHOENIX CONTACT										
+	3273110	PTFIX 6/18X2,5–NS35 GY – Distribution block							PHOENIX CONTACT										
A0	6AG1155–6AU01–7BN0	ET 200SP, PROFINET interface module IM 155–6PN Standard							SIEMENS										
A0	6AG1193–6AR00–7AA0	ET 200SP BA 2xRJ45							SIEMENS										
A1	6AG1134–6GF00–7AA1	AI 8xl 2–/4–wire BA							SIEMENS										
A1	6AG1193–6BP00–7DA0	ET 200SP BU15–P16+A0+2D push–in terminals							SIEMENS										
A2	6AG1134–6GF00–7AA1	AI 8xl 2–/4–wire BA							SIEMENS										
A2	6AG1193–6BP00–7BA0	ET 200SP BU15–P16+A0+2B push–in terminals							SIEMENS										
A3	6AG1135–6HD00–7BA1	ET 200SP AQ 4xU/I Standard							SIEMENS										
A3	6AG1193–6BP00–7DA0	ET 200SP BU15–P16+A0+2D push–in terminals							SIEMENS										
A4	6AG1131–6BH01–7BA0	ET 200SP DI 16x24 V DC standard							SIEMENS										
A4	6AG1193–6BP00–7DA0	ET 200SP BU15–P16+A0+2D push–in terminals							SIEMENS										
A5	6AG1132–6GD51–7BA0	ET 200SP RQ 4x24VDC/2A CO ST							SIEMENS										
A5	6AG1193–6BP00–7DA0	ET 200SP BU15–P16+A0+2D push–in terminals							SIEMENS										
F00	952110	DG M TT 2P 275 – SPD							DEHN										
F1	2302038	Fuseholder for cylindrical fuses – 2P for fuses 10,3x38mm – BCH 2x38 32A 690V							ITALWEBER										
F11	TBS4F	MODULAR TERMINAL WITH FUSE							CEMBRE										
F12	TBS4F	MODULAR TERMINAL WITH FUSE							CEMBRE										
F13	TBS4F	MODULAR TERMINAL WITH FUSE							CEMBRE										
F2	TBS4F	MODULAR TERMINAL WITH FUSE							CEMBRE										
F3	TBS4F	MODULAR TERMINAL WITH FUSE							CEMBRE										
F4	TBS4F	MODULAR TERMINAL WITH FUSE							CEMBRE										
F5	TBS4F	MODULAR TERMINAL WITH FUSE							CEMBRE										
F6	TBS4F	MODULAR TERMINAL WITH FUSE							CEMBRE										
FG1	2301138	Fuseholder for cylindrical fuses – 1P+N for fuses 10,3x38mm – BCH 1x38N 32A 690V							ITALWEBER										
FN1	TBS4F	MODULAR TERMINAL WITH FUSE							CEMBRE										
FRK0	TBS4F	MODULAR TERMINAL WITH FUSE							CEMBRE										
G1	2903149	TRIO–PS–2G/1AC/24DC/10 – Power supply unit							PHOENIX CONTACT										
N1	D5020S	I.S. SIL2 HART Isolating Driver							SIEMENS										
P–PN–0T331.1	EJB–H	ENCLOSURE 832x612x412 mm L x H x D							CMC										
Q0	PSRC–320	SELECTOR SWITCH							CMC										
UHS–3303A/B	PSRC–D220	CHANGE OVER SWITCH							CMC										
XL–0T331–AL	PLD20	PILOT LAMP RED							CMC										
XL–0T331–AL	O2OPD9	LAMPHOLDER BA9S, DIRECT SUPPLY							ELFI										

General Contractor :		Manufacture : 	Project : P–PN–0T331.1 AMMINE COLUMNS	Customer : 	Object :			
							By :	
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					Rev. 1	Data : 27/02/2026	Sheet : 13	of : 14

